

"Chapter-8"

Data compression

process of reducing the amount of data required to represent a given quantity of information

Data $\neq$ Information

Data redundancy

irrelevant or repeated information.

reasons for compression

- (i) Transmission.
- (ii) Storage.

Ex-1

SD TV 2 hrs color sequence

Size 720×480

30 frames/sec

$$\text{Size} = 30 \times (720 \times 480) \times 3 \text{ bytes/pixel}$$

$$\approx 31 \text{ MB}$$

$$\therefore 31104000 \times 60 \times \frac{\text{sec}}{\text{hr}} \times 2 \text{ hr}$$

$$\approx 2.24 \times 10^8 \text{ bytes}$$

$$= 224 \text{ MB}$$

$$C_R = \frac{n_1}{n_2} \quad [\text{compression ratio}]$$

n_1 and $n_2 \rightarrow$ no. of info carrying in 2 data sets that represent same information.

$$R_D = 1 - \frac{1}{C_R}$$

Coding redundancy

Codeword / sequence of symbols represent a piece of information. Number of symbols required is its length. Minimize codeword. Use variable length codeword.

From histogram calculations

$$p(r_k) = \frac{h(r_k)}{n} = \frac{n_k}{n}$$

$$L_{av} = \sum_{k=0}^{L-1} l(r_k) (p(r_k))$$

$$C = \frac{M \times N \times 8 \text{ bit}}{M \times 8 \times L_{av}}$$
$$= \frac{8}{L_{av}}$$

$$M N L_{av} = M N \sum_{k=0}^{L-1} l(r_k) p(r_k)$$

Ex-1

Table

r_k	$P_r(r_k)$	Code1	$l_1(r_k)$	Code2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	10000000	8	1	1
$r_{186} = 186$	0.25	11000100	8	000	3
$r_{255} = 255$	0.03	11111111	8	001	3
r_0 (anything else)	0	—	8	—	0

$$\therefore (2 \times 0.25) + (1 \times 0.47) + (3 \times 0.25) + (3 \times 0.03)$$

$$= 1.81 \text{ bits}$$

$$C = \frac{256 \times 256 \times 8}{256 \times 256 \times 1.81}$$

$$= 4.4199$$

$$R = 1 - \frac{1}{4.4199} = 0.77375$$

$\approx 77.4\%$ redundancy.

Spatial redundancy

Intensities of each pixel may correlate to its neighbours. Many info is replicated unnecessarily.

Histogram equiprobable

pixels are independent vertically

pixels are maximally correlated horizontally.

variable length coding is not possible.

Run length pair

original bit $256 \times 256 \times 8$

$$\text{run length} = (256 + 256) \times 8$$

$$\therefore \text{Compression ratio} = \frac{256 \times 256 \times 8}{(256 + 256) \times 8}$$
$$= 128 : 1$$

NOTE

Many images has info. that are ignored by human visual system. Remove these extraneous data.

Information theory

$$I(E) = \log \frac{1}{P(E)}$$

$$= -\log P(E).$$

Average information per event or the
entropy of the source

$$H = \sum_{k=1}^{L-1} -P(a_k) \log P(a_k)$$

Entropy is the measurement of the
average information in an image.

NOTE

It is not possible to encode input image with fewer than H bits/pixel.

Ex-1

r_k	$P_r(r_k)$
$r_{87} = 87$	0.25
$r_{128} = 128$	0.47
$r_{186} = 186$	0.25
$r_{255} = 255$	0.03
$k \neq 87, 128, 186, 255$	0

$$\tilde{H} = \sum_{k=1}^{L-1} -P_r(r_k) \log P(r_k)$$

$$= -[0.25 \times \log_2 0.25 + 0.47 \log_2 0.47 + 0.25 \log_2 0.25 + 0.03 \log_2 0.03]$$

$$= -[-\frac{1}{2} - 0.511956 + \frac{1}{2} - 0.151768]$$

$$= 1.664 \text{ bits/pixel}$$

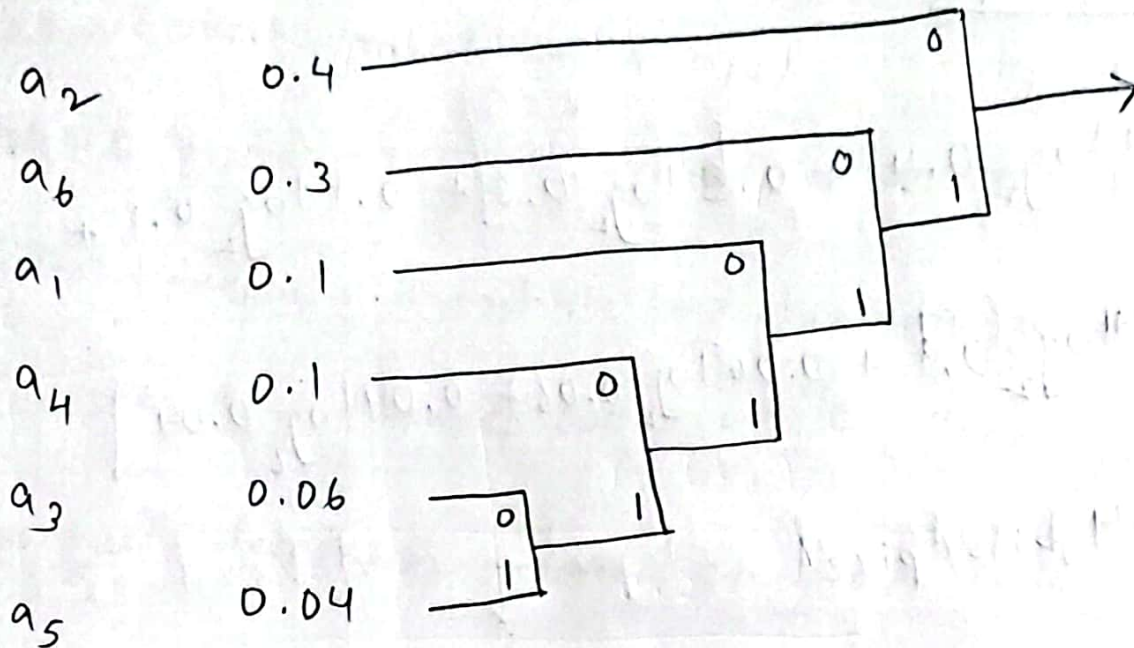
Amount of entropy is not the same as the information in an image.

Huffman coding

Ex-1

S_c	P_r	Sort(P_r)	merge1	merge2	merge3	merge4
a_1	0.1	a_2 0.4(1)	0.4(1)	0.4(1)	0.4(1)	0.6
a_2	0.4	a_6 0.3(00)	0.3(00)	0.3(00)	0.3(00)	0.4
a_3	0.06	a_1 0.1(011)	0.1(011)	0.2(010)		
a_4	0.1	a_4 0.1(0100)	0.1(0100)	0.1(011)	0.3(01)	
a_5	0.04	a_3 0.06(01010)	0.1(0101)			
a_6	0.3	a_5 0.04(01011)				

Easy method



$a_2 \rightarrow 0$

$a_6 \rightarrow 10$

$a_1 \rightarrow 110$

$a_4 \rightarrow 1110$

$a_3 \rightarrow 11110$

$a_5 \rightarrow 11111$

$$\begin{aligned} L_{avg} &= (0.4 \times 1) + (0.3 \times 2) + \\ &\quad (0.1 \times 3) + (0.1 \times 4) + \\ &\quad (0.06 \times 5) + (0.04 \times 5) \\ &= 2.2 \text{ bits/symbol} \end{aligned}$$

Original image = 3 bit

$$\begin{aligned} \therefore C_r &= \frac{3}{2.2} \\ &= 1.3636 \end{aligned}$$

Calc entropy

$$-\left[0.4 \log_2 0.4 + 0.3 \log_2 0.3 + 0.1 \log_2 0.1 + 0.11 \log_2 0.1 + 0.06 \log_2 0.06 + 0.04 \log_2 0.04\right]$$

$$= 2.14 \text{ bits/pixel}$$

no info is lost.

$$R_D = 1 - \frac{1}{1.3636}$$

$$= 26.7\%$$

NOTE

codes one symbol at a time.

Block code $\rightarrow$ each symbol is mapped to a block code.

Instantaneously decodable $\rightarrow$ does not need to foresee the future codes while decoding.

Uniquely decodable $\rightarrow$ any string of code symbols can be decoded in only one way.

Hollomb Coding

Combination of unary code of quotient, $\text{floor}[n/m]$ and the binary representation of remainder $(n \bmod m)$

Can only represent non-negative integer inputs.

with exponentially decaying probability distributions.

Optimally encoded.

Ex. 1

$$n = 9$$

$$m = 4$$

$$q = \left\lfloor \frac{9}{4} \right\rfloor$$

$$\text{Find } G_4(9) = 2$$

Unary code of integer q is defined as

1s followed by a 0

Since 2 so 110.

now

$$k = \log_2 m$$

$$= \log_2 4$$

$$= 2$$

$$c = 2^2 - 4$$

$$= 0$$

$$r = 9 \bmod 4$$

$$= 1$$

$$\text{if } r \geq c$$

$$r' = 01 [r+c \text{ in } k \text{ bits}]$$

$$\therefore \text{final result} = 11001$$

Ex-2

$$h_4(7)$$

$$m = 4$$

$$n = 7$$

$$q = \frac{7}{4} \\ = 1$$

$$\therefore 10$$

now

$$k = \log_2 4 \\ = 2$$

$$c = 2^k - m$$

$$= 2^2 - 4$$

$$= 0$$

$$r = 7 \bmod 4 = 3$$

$$r \geq c$$

$$r' = 011 [r+c \text{ in } k \text{ bits}]$$

$$\therefore 1011$$

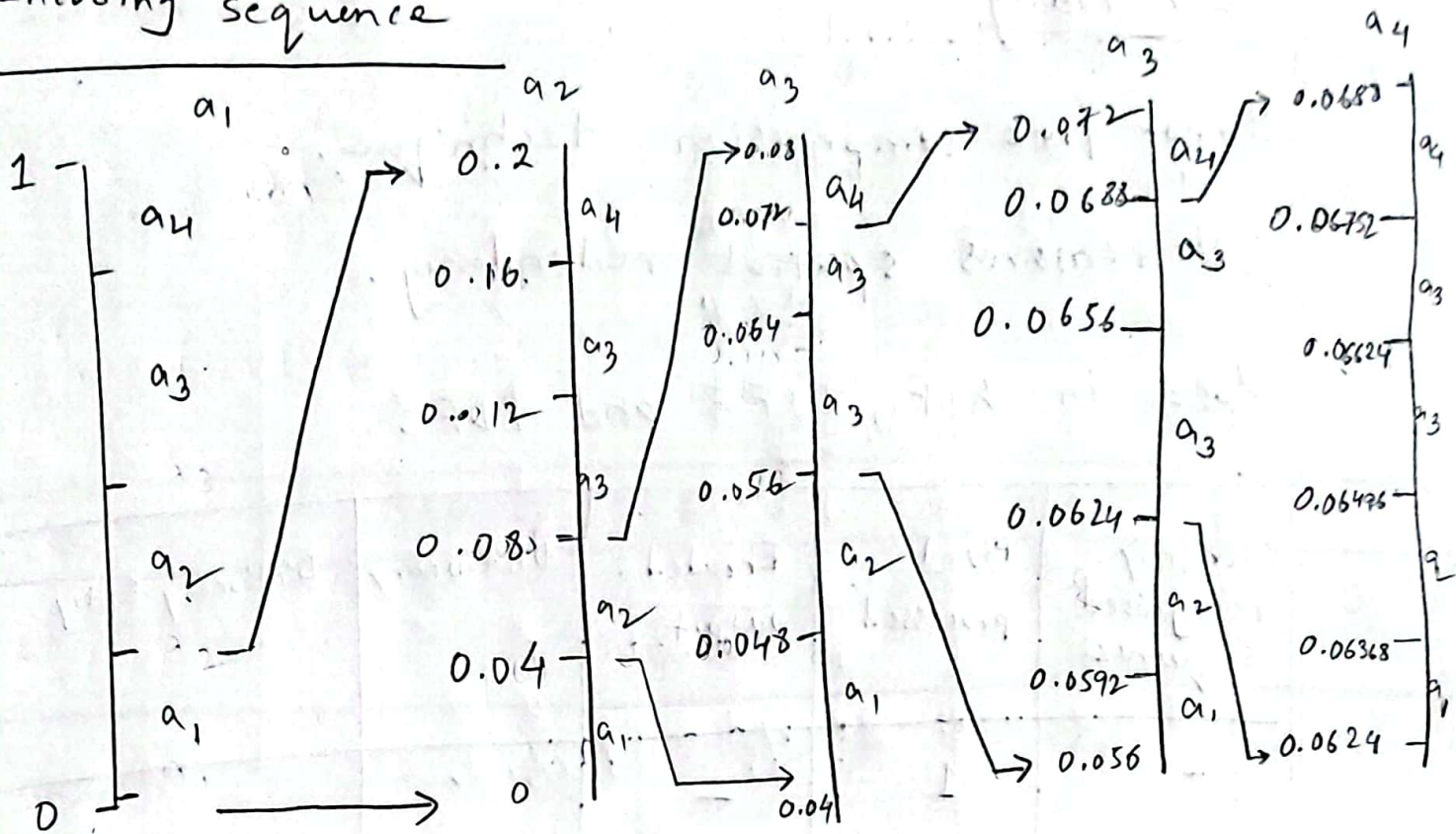
Arithmetic coding

One to one correspondence between symbol and code doesnot exist.

Ex-1

s_c	P_r	Subinterval
a_1	0.2	$[0.0, 0.2)$
a_2	0.2	$[0.2, 0.4)$
a_3	0.4	$[0.4, 0.8)$
a_4	0.2	$[0.8, 1.0]$

Encoding sequence



$\therefore$ any value between 0.0688 and 0.06752

is the answer. So let's say 0.068 (Ans.).

LZW coding

Error free compression technique.

It removes spatial redundancy.

Used in GIF, TIFF and PDF.

Currently recognised sequence	Pixel processed	Encoded output	Dictionary	Dictionary Entry
-	1	-		
1	2	1	8	1-2
2	3	2	9	2-3
3	4	3	10	3-4
4	1	4	11	4-1
1	2	-	-	1-2
1-2	3	8	12	1-2-3
3	4	10	-	3-4-1
3-4	1	10	13	3-4-1
1	2	-	-	1-2-3

Image

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

0 - 0

⋮ ⋮

7 7

8 - 1-2

CRS	PP	EO	D	DE
1-2	3	-	-	1-2-3
1-2-3	4	12	14	1-2-3-4
4	1	-	-	4-1
4-1	2	11	15	4-1-2
2-	3	-	-	2-3
2-3	4	9	16	2-3-4
4	-	4	-	-

∴ 2, 3, 4, 3, 10, 12, 11, 9, 4

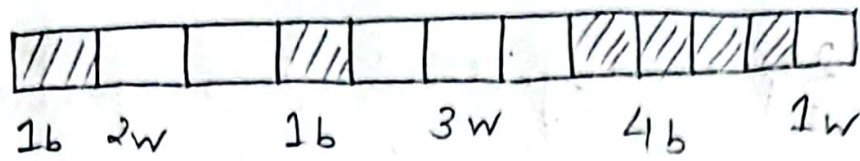
$$\therefore C_R = \frac{16 \times 3}{9 \times 4}$$
$$= 1.33$$

$$R_D = 1 - \frac{1}{1.33}$$
$$= 25\% \quad (\text{Ans.})$$

Run length encoding

It is a mapping scheme.

Eliminates small spatial redundancies.



Uses run length pairs

$(0, 1), (1, 2), (0, 1), (1, 3)$

$\downarrow$ $\searrow$
 black no.
 of
 pixels

Ex-1

000001100000100000000101100000

0525181125 : grey level of the first run
is '0' (black)

Ex-2

0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	1	1	0	0
0	1	1	1	0	0	1	0
0	0	1	0	0	1	1	0
1	1	1	1	0	1	0	0
0	0	0	0	0	0	0	0

1st row : 8

2nd row : 0, 4, 4

3rd row : 1, 2, 5

4th row : 1, 5, 2

5th row : 1, 3, 2, 1, 1

6th row : 2, 1, 2, 2, 1

7th row : 0, 4, 1, 1, 2

8th row : 8

Encoded mode

(x, y)



run
length



color index

Absolute mode

(x, y)



always
0

0 → end of line

1 → end of image

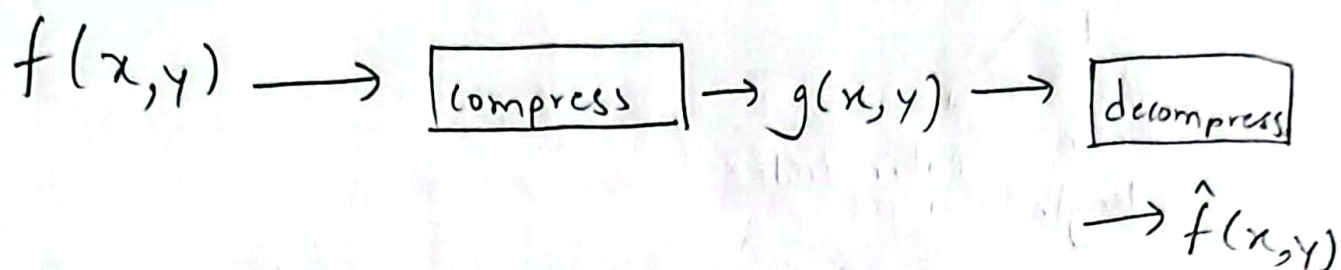
2 → move to new position

3-255 → Specify pixels individually.

Fidelity criterion

The degree of exactness with which something is copied or reproduced is called fidelity.

2 types objective & subjective



$$\hat{f}(x, y) = f(x, y) + \underbrace{e(x, y)}_{\text{error}}$$

Subjective



based on
human observers

Showing a decompressed
image to a viewer and
avg their evaluations.

Objective



mathematically defined
criteria.

When the level of info
loss can be expressed as
a function of the original /
input image and the compressed
and subsequently decompressed
output image is said to be
based on an objective fidelity criteria.

$$e(x, y) = \hat{f}(x, y) - f(x, y)$$

↓
error

Root Mean Squared error

$$e_{rms} = \sqrt{\frac{1}{MN} [\hat{f}(x, y) - f(x, y)]^2}$$

Signal to Noise Ratio

$$SNR_{ms} = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2}$$

$$\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2$$

↑ SNR means better image.

————— END of chap 8 —————